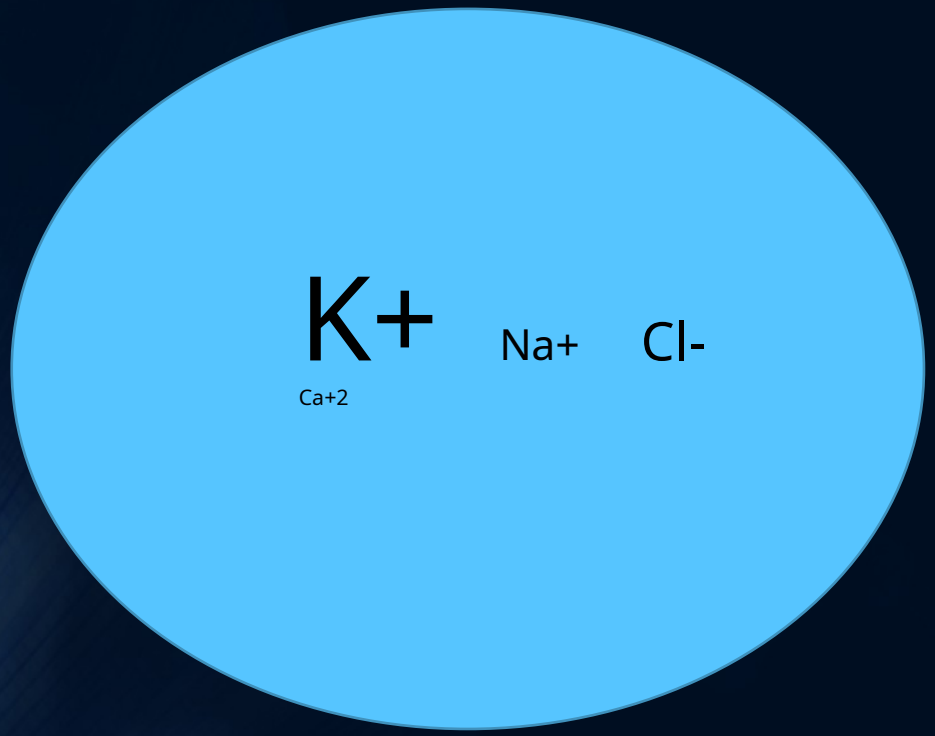
Renal Physiology

REGULATION OF ELECTROLYTES

Normal Electrolyte Values



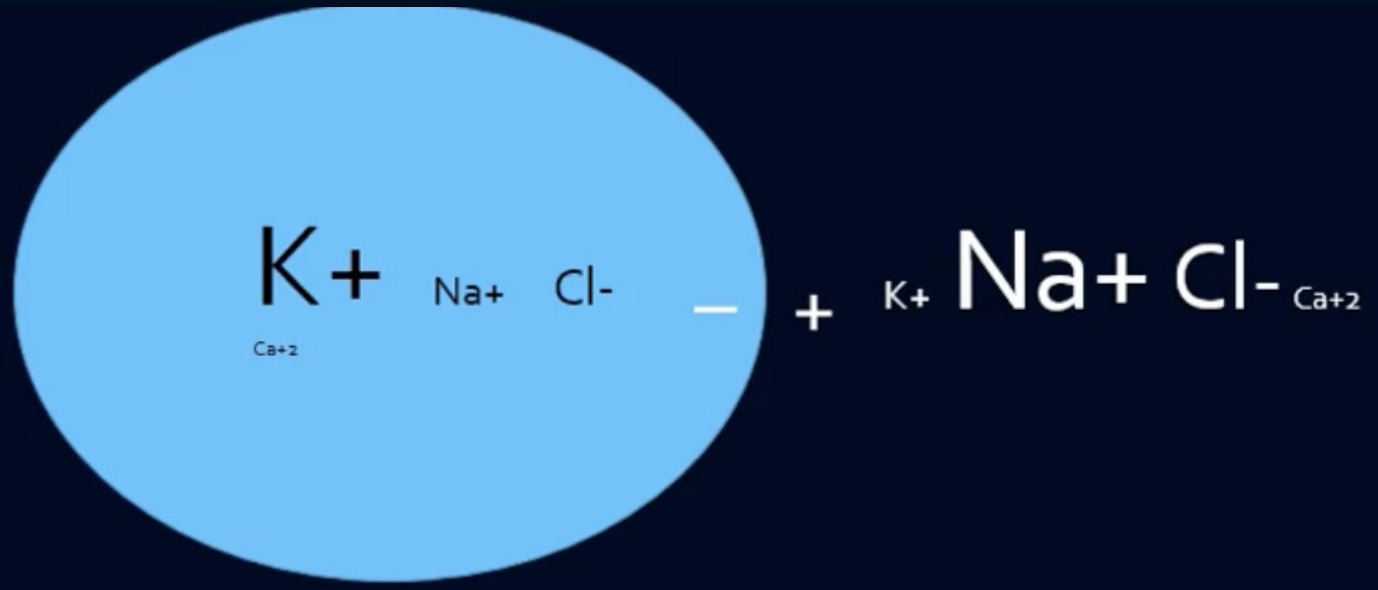
K+ Na+ Cl- Ca+2

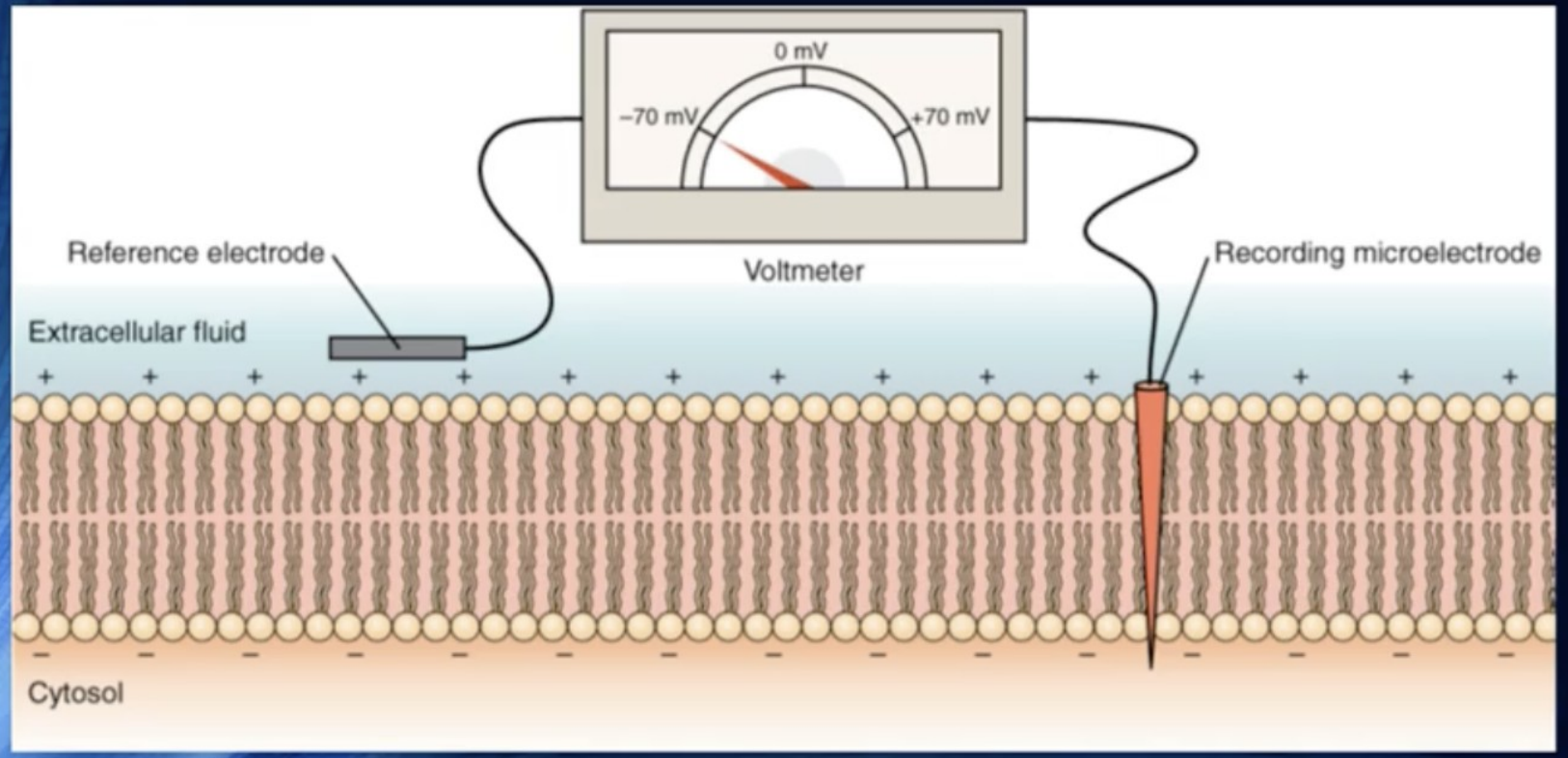
Normal Concentrations of Solutes in Fluid Compartments

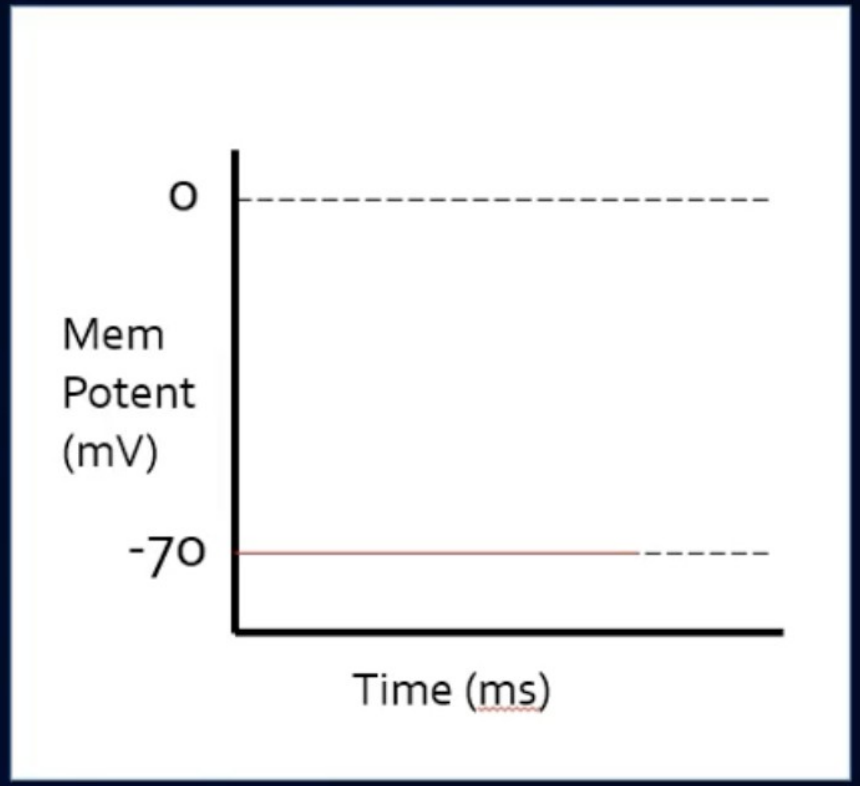
- ICF vs ECF electrolytes
  - Na+/K+ pump
  - Properties of plasma membrane (leaky channels etc.)
- Plasma vs interstitial electrolytes
- Protein concentrations – why are plasma and interstitial so different?
- These numbers will vary from source to source

	ECF		ICF
	Plasma	**Interstitial**	
Na+	135 – 142	140	15
K+	3.5 – 5.1	4.5	120
Ca+2	1.3	1.2	0
Cl-	100	110	20
HCO3-	22 – 26	25	15
Protein	7	<1	30
Total	300	300	300

Resting Membrane Potential







Renal Physiology

REGULATION OF ELECTROLYTES - NA+

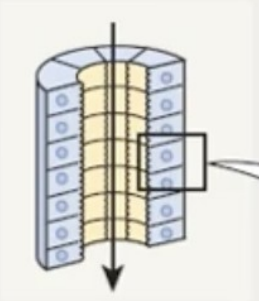
Regulation of Plasma Na+

- Normal range of Na+ varies by tissue but 140 is a good number to use for plasma
- Na+ concentration regulated through altering filtration and reabsorption of Na+ by the kidneys
- Hormones that regulate plasma Na+
  - Aldosterone*
  - Atrial Natriuretic Peptide

	ECF		ICF
	Plasma	**Interstitial**	
Na+	135 – 142	140	15
K+	3.5 – 5.1	4.5	120
Ca+2	1.3	1.2	0
Cl-	100	110	20
HCO3-	22 – 26	25	15
Protein	7	<1	30
Total	300	300	300

Aldosterone

- Stimuli
  - Hyperkalemia
  - Increased Angiotensin II secretion
- Actions
  - Increases probability that ENaC and ROMK channels will be open on apical side
  - Increases expression of ENaC and ROMK channels, Na+/K+pump
  - Increases rate of catalysis by Na+/K+ pump
- Influence of Fluid Balance and Osmolarity
  - K+ is secreted into lumen; Na+ is reabsorbed
  - Net increase in ion reabsorption causes increase in H2O reabsorption



Lumen of distal nephron

P cell of distal nephron

Interstitial fluid

Blood

2 Hormone-receptor complex initiates transcription in the nucleus.

1 Aldosterone combines with a cytoplasmic receptor.

Transcription

2

mRNA

Aldosterone

3 Translation and protein synthesis makes new protein channels and pumps.

Aldosterone receptor

New channels

New pumps

ATP

4 Aldosterone-induced proteins modulate existing channels and pumps.

K^+ secreted

K^+

K^+

ATP

K^+

Na^+ reabsorbed

Na^+

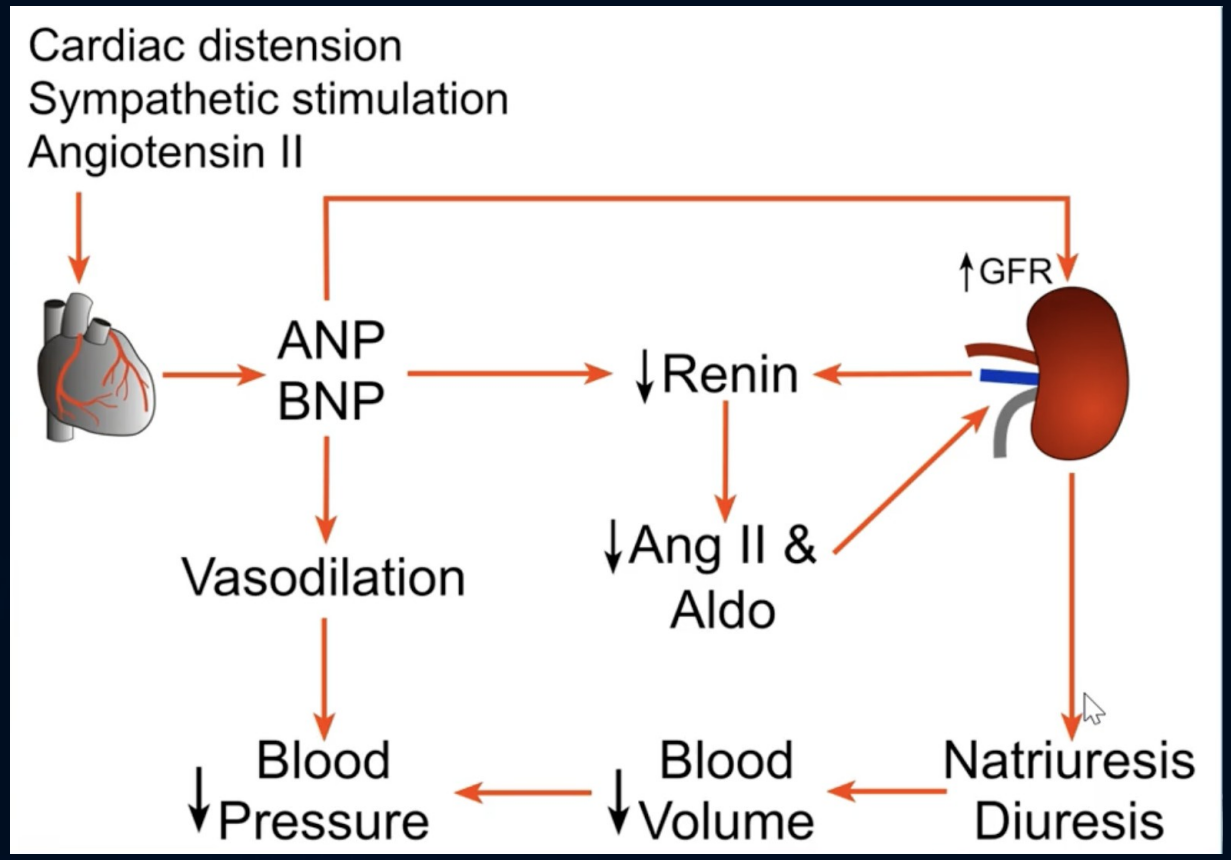
Na^+

5 Result is increased Na^+ reabsorption and K^+ secretion.

Na^+

Atrial Natriuretic Peptide

- Stimuli
  - Stretch of specialized stretch receptors in atrial myocytes
- Actions
  - Dilates afferent arteriole – increased GFR thus more Na+ filtered
  - Suppresses release of renin – dec. Ang II thus dec. ALDO
  - Decreases probability of ENaC being open in DCT and CD (due to dec. ALDO)
  - Decreases rate of activity Na+/K+ pump in DCT and CD (due to dec. ALDO)



Renal Physiology

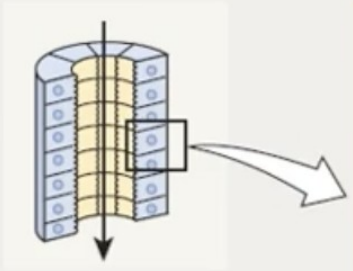
REGULATION OF ELECTROLYTES - K+

Regulation of Plasma K+

- Normal range of K+ varies by tissue but 3.5-5.0mEq/L is a good number to use for plasma
- K+ concentration regulated through altering secretion of K+ by the kidneys
- Hormones that regulate plasma Na+
  - Aldosterone*

	ECF		ICF
	Plasma	**Interstitial**	
Na+	135 – 142	140	15
K+	3.5 – 5.1	4.5	120
Ca+2	1.3	1.2	0
Cl-	100	110	20
HCO3-	22 – 26	25	15
Protein	7	<1	30
Total	300	300	300

Regulation of plasma K+ by Aldosterone



Lumen of distal nephron

P cell of distal nephron

Interstitial fluid

Blood

2 Hormone-receptor complex initiates transcription in the nucleus.

1 Aldosterone combines with a cytoplasmic receptor.

Transcription

2 mRNA

3 Translation and protein synthesis makes new protein channels and pumps.

Aldosterone

Aldosterone receptor

New channels

New pumps

ATP

4 Aldosterone-induced proteins modulate existing channels and pumps.

K^+ secreted

K^+

K^+

K^+

ATP

Na^+ reabsorbed

Na^+

Na^+

Na^+

5 Result is increased Na^+ reabsorption and K^+ secretion.

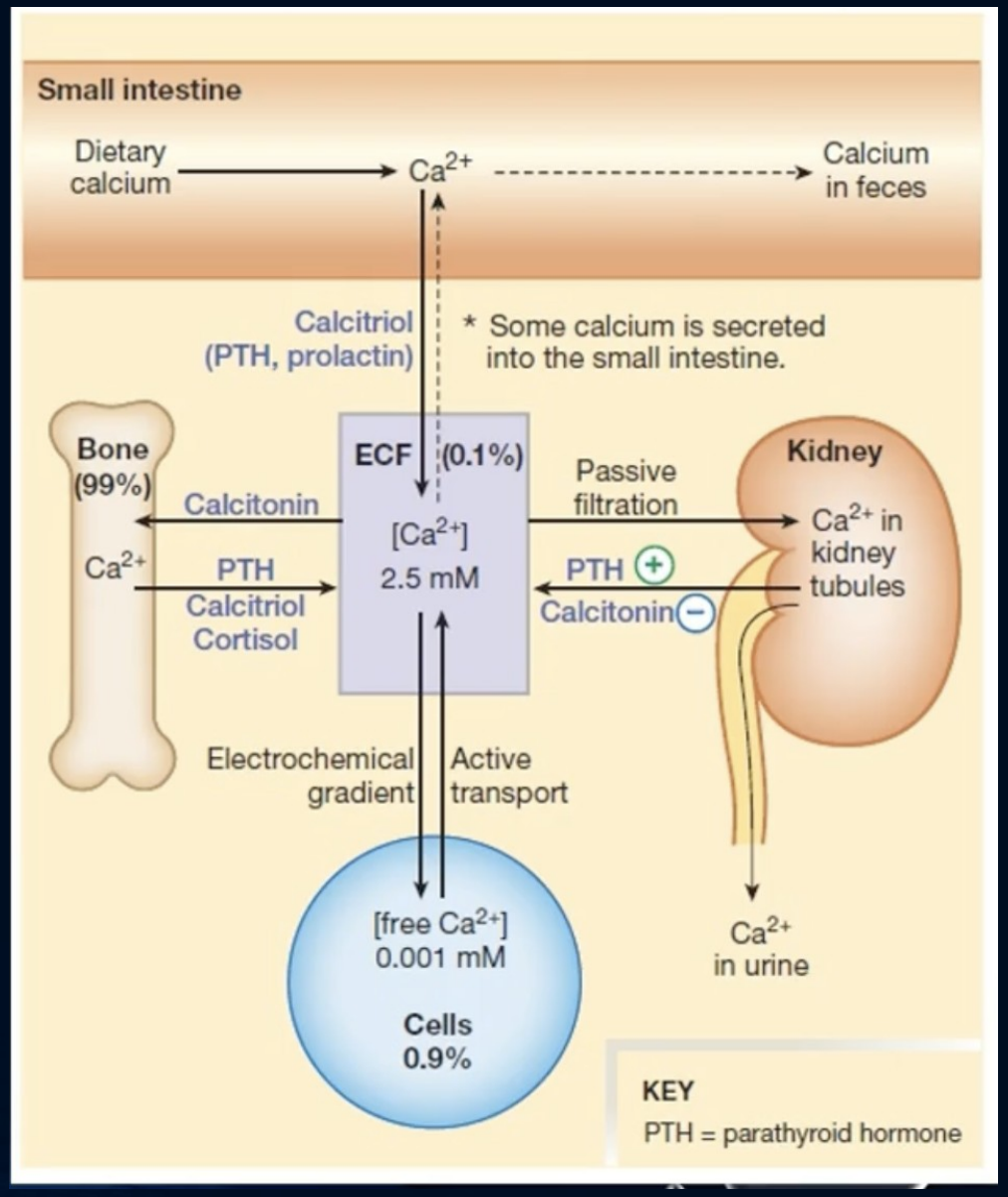
Renal Physiology

REGULATION OF ELECTROLYTES – CA2+

Regulation of Plasma ca2+

- Normal range of Ca2+ varies by tissue but 4.3-5.3mEq/L is a good number to use for plasma
- Ca2+ concentration regulated through altering the following:
  - Absorption by the GI tract
  - Reabsorption by the kidneys
  - Bone mineralization / demineralization
- Hormones that regulate plasma Na+
  - Parathyroid hormone
  - Vitamin D
  - Calcitonin

	ECF		ICF
	Plasma	**Interstitial**	
Na+	135 – 142	140	15
K+	3.5 – 5.1	4.5	120
Ca+2	1.3	1.2	0
Cl-	100	110	20
HCO3-	22 – 26	25	15
Protein	7	<1	30
Total	300	300	300

Regulation of ECF Ca2+

- Parathyroid hormone (stim is low ECF Ca2+
  - Raise ECF Ca2+ by increasing bone reabsorption, Ca2+ absorption and Ca2+ reabsorption
- Vitamin D
  - Calcitriol (active form) increases Ca2+ absorption raising ECF Ca2+
- Calcitonin
  - Decreases ECF Ca2+ by increasing bone matrix deposition (inhibits osteoclasts)
  - Inhibits Ca2+ reabsorption by the kidneys thereby decreasing ECF Ca2+